CoRe: Supplementary Technical Report

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Abstract

This report provides the technical specification and extended evidence for CoRe (**C**leaning **o**r **Re**calling), the evaluation protocol introduced in *Cleaning or Recalling? Knowledge Dependence in LLM-based Data Cleaning*. We describe controlled-table construction and acceptance, archived execution and scoring, Rayyan retrieval and repair-credit criteria, adaptation to other benchmarks, offline verification routes, extended displays, and sensitivity checks on the existing evidence. We specify the continuation screen, cell-level metrics and statistical comparisons, distinguishing construction-seed ranges from fixed-construction intervals and conditional field-check uncertainty.

The seven numbered sections correspond to the manuscript's Report §1–§7 links. Tables and figures in this report carry an S prefix; unprefixed numbers refer to the main paper. Code and data links point to hinsonver/CoRe-Benchmark.

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1 Controlled-table construction and acceptance

1.1 Construction contract

O is the original benchmark. W and S retain its rows, error cells, chunks and non-identity values, and transfer errors to one-to-one renamed identity values. W uses English words and S uses syllables. F uses other real records with the same schema and error recipes or formatting rules. Its error counts can differ when recipes cannot transfer or a formatting rule applies to more values. O/W/S support paired comparisons; F also changes the records and their repair difficulty, as described in main-paper Section 3.1.

The construction produces 21 new tables: three conditions (F/W/S) at three seeds for each of Hospital and Movies, plus three Beers conditions at one seed, or $9 + 9 + 3$. Together with three original-condition tables, these give 24 cleaning inputs. Conflict inputs and the Rayyan audit are counted separately. The surrogates contain about 41,000 fictitious values. Main sources are code/src/validate_tables.py, the data/final/ materials, and the paper's paper/latex/tables/construction.tex, paper/latex/tables/movieserrors.tex and paper/latex/tables/checks.tex.

1.2 Hospital

The original has 1,000 quality-measure rows for 45 hospitals and 509 erroneous cells. Of these, 508 replace a character by `x`; in 96% of them every occurrence of that character is replaced. Column error rates range from 2.3% to 3.9%.

F uses the July 2026 CMS registry. Quality measures are joined with hospital information and converted to the benchmark's 20 columns and formats: lower case, digits-only phone numbers and provider numbers without leading zeros. Whole hospitals are added in provider order, starting with those covered by the benchmark, until 1,000 rows are filled. The table has 34 hospitals, of which 33 occur in O. All quality measures have changed, so these are new rows about largely the same entities. Each column receives the original number of errors, with character substitutions drawn by the original position-weighted rule at newly drawn error locations.

The 296 distinct identity values each map to one fictitious value. S uses 24 syllables and W uses 63 English words, with generic suffixes such as `regional hospital` and `street`. Provider numbers, ZIP codes and phone numbers become random digits. At each original erroneous position, the transferred corruption replaces a character chosen by position with `x`. The small syllable inventory is also the inventory used by the decoding analysis in main-paper Section 8.3; it was not supplied to the cleaner.

Table relationships and geographic meaning. Column-wise bijections preserve repeated-value relationships and equality-based dependencies. All non-identity values, including State, remain unchanged. ZIP codes and phones are generated independently of those states, so the replacements do not preserve their geographic meaning. For example, the first W reference row pairs State `al` with ZIP `50438`, which the USPS labeling list places in Garner, Iowa. The acceptance checks cover structure, error profiles and naming; geographic validity is not one of their criteria.

An independent edit audit parses 80 archived Hospital runs from the eight cleaners: O and three seeds each of F/W/S, comprising 4,000 chunks. Its 48,000 integer count comparisons agree with the saved per-chunk counts. Table S1 summarizes the geographic fields in the 48 W/S runs. Each entry counts edits across model/construction runs, not distinct hospitals. State is outside the identity-column score: its 130 clean-cell edits and eight unsuccessful error edits do not directly enter identity F1.

Table S1: Geographic-field edits across the 48 Hospital surrogate runs. Wrong error edits change an erroneous cell without reaching its reference; clean-cell edits change a reference-consistent cell.

Field	Edits	Credited repairs	Wrong error edits	Clean-cell edits
Address	898	492	277	129
City	1,601	530	146	925
County	1,231	602	112	517
Phone	1,133	32	890	211
State	1,089	951	8	130
ZIP	1,121	73	835	213

Two uncredited identity edits reproduce the original row's value: Qwen2.5-14B writes `Anchorage` for a correct

W city, and Qwen2.5-32B writes `Lauderdale` for a correct S county. These matches and other geographic edits do not identify why the cleaner made them or isolate the effect of geographic conflicts. The Hospital contrasts therefore describe joint entity replacements, including changes in factual associations. The audit's inputs, edit records and command are specified in Section 7.8.

1.3 Movies

The 1,000-row original subset comprises all 190 rows replaced wholesale by Rotten Tomatoes records and 810 other rows sampled without replacement using Python seed 0. The selected rows retain benchmark order and are restricted to ten columns that IMDb public datasets can fill. Two table-wide formatting rules apply in addition. The 1,965 error cells include 379 identity errors. The column breakdown is in `tables/movieserrors.tex`.

For example, *American Ultra* (2015) has Rotten Tomatoes slug `american_ultra` in place of IMDb identifier `tt3316948`, year range `2014 2015 2016`, and Rotten Tomatoes score pair `5.4/10,6/10` in place of IMDb rating 6.3. Credits can add, omit or respell names. These are source differences, not uniformly typographical errors.

1.3.1 Fresh film selection and formatting

For each original film, the constructor selects an absent film from the September 2026 IMDb snapshot. Candidates are feature films with runtime, genres and at least 1,000 votes, released in the same year or, if necessary, within two years. Selection minimizes log-vote-count distance. Clean rows use the first two directors and writers, the first three distinct actors, and durations such as `99 min`.

Fresh row i corresponds to original row i for applying the 190 source-style error recipes. Slugs, year ranges, durations and ratings are instantiated from the original patterns, and the formatting rules are applied to the new values.

- People recipes record names kept, dropped, respelled or added. Added names come from the fresh film's other columns or other IMDb credits, mirroring the additional names in the other source.
- Title recipes use the shortest sequence of one or two edits from eleven kinds: capitalization, `&`/`and`, accents, punctuation, or added/dropped subtitles, tokens and articles. The error-profile table separately records 12 title edit recipes.
- Genre recipes preserve the list slots and map the fresh film's genres to Rotten Tomatoes vocabulary, for example `Action & Adventure,Comedy` for `Action,Comedy`.
- A recipe that cannot apply to its assigned fresh cell moves to another source-style row able to take it.

Four director and four writer recipes remain inapplicable after relocation. F therefore has 371 identity errors, compared with 379 in O/W/S. At seed 0, F has 2,195 errors overall. Every fresh film has at least 1,000 votes, so the separator rule affects 1,000 rating-count cells rather than 768. These counts describe different fresh records under the same mechanisms, not paired error cells.

1.3.2 Surrogate names and source-style errors

The surrogates rename 6,018 values one to one, including 4,018 people. W person names combine English words into first and compound surnames; S person names use two syllable words. Titles are generated independently of the real title, rather than retaining its function words, numbers and punctuation. Generic shapes include `The X`, `X of the Y`, `X: Y Z` and `X 2`, sampled at the measured rates: 18.5% initial `The`, 6.1% colon and 2.9% final number. W excludes all words appearing in a real title in the table.

Each transferred title error requires a compatible generated shape, such as an `and` token or a subtitle. Identifiers become unique fictitious IMDb numbers; the slug is rebuilt from the dirty fictitious title in the form of the original slug. This preserves the corruption mechanism without providing a mapping from a fictitious slug to a random reference identifier.

1.3.3 Recovery evidence

The 105 hour/minute duration errors convert to the reference in 50 cases but not in 55; another 80 duration errors are empty and two are other source differences. A format conversion therefore does not always recover the preferred source value. For years, the middle candidate matches 167 of the 190 references. Main-paper Section 7 discusses this candidate-rule example.

1.4 Beers

The 1,000-row subset includes all 127 rows with city errors and 873 others sampled without replacement using Python seed 0. Selected rows retain benchmark order. This subset contains 421 breweries, rather than 1,000 independent entities. Its 565 errors comprise 311 percentage suffixes and 254 location cells: a state glued to a city and an empty state cell in each of 127 rows. Correcting the suffix or splitting the location retains the necessary information in the input.

F uses a September 2026 Wikidata query for US beers with alcohol content, style and brewery location. Original beers are excluded, no brewery contributes more than 25 beers, and 1,000 beers from 94 breweries are selected. Corruptions retain the original counts and rendering, including the floating-point representation of percentage values. The original has 421 breweries; F is descriptive in the knowledge analysis because brewery concentration and alcohol-content distributions differ.

Surrogates keep generic words such as `Brewing`, `Company`, `IPA`, `Stout` and `San`, and years. Other words become W-pool or S-syllable words. Numbers retain their lengths; apostrophes remain only for possessives and `O'`. Repeated real values map consistently, including a brewery named for a city. A one-word city remains one word, so a correct generated city does not resemble the city/state-gluing error being tested.

1.5 Acceptance and construction seeds

The six check categories contain 67 structural, 137 error-kind/count, 15 error-strength, 263 naming/leakage, 6 fresh-entity and 18 reproducibility checks, totalling 506. Structural checks distinguish paired surrogates from fresh records. Movies column counts generally have an 85%–115% tolerance, with error-kind constraints; rating-count errors are checked against the formatting rule. Naming checks include bijection, no original value or title-word reuse, pool membership and title-shape similarity against random pairing.

Eighteen tables are rebuilt under two hash seeds and must match stored files byte for byte. The three main fresh tables use saved source snapshots. The experimental tables pass 505 checks. Hospital W at seed 0 has one/two/at-least-three-character replacement shares of 62.1%/31.5%/6.4%, versus 53.9%/29.2%/16.9% in O. Its largest difference is 10.5 percentage points, 2.5 points beyond the eight-point tolerance. Seeds 1 and 2 pass this check. The error positions remain fixed in all three seeds. The archived sensitivity checks (Section 7) give the distributions and condition contrasts separately for the passing seeds. Character-replacement counts describe the corruption profile, rather than a measured model-difficulty scale.

Across construction seeds, surrogates redraw names and applicable transferred-error variants. Fresh tables retain the same clean records and redraw errors and their variants. Seed ranges concern naming/error draws; cluster intervals describe variation across entities and replies within one fixed construction.

1.6 Protocol and construction illustrations

Figures S1 and S2 summarize the protocol and its paired construction example.

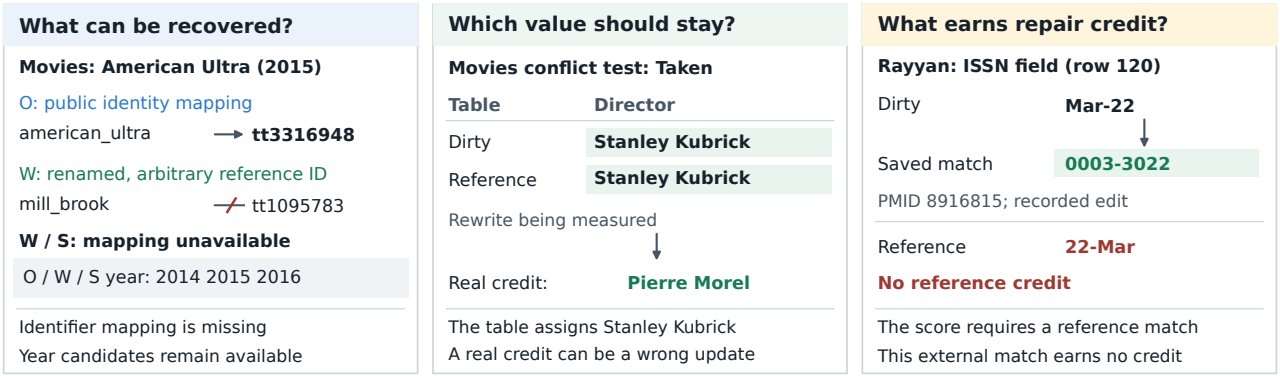


Figure S1: Three complementary questions for interpreting repair scores: recovery evidence, preservation of assigned values, and reference credit. The panels illustrate separate tests, not a shared sample.

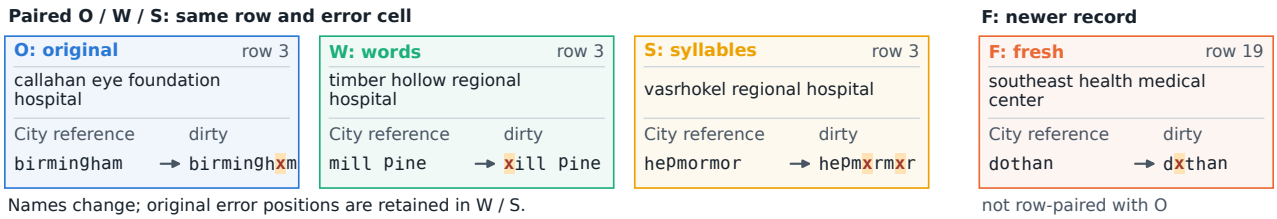


Figure S2: Hospital construction example. O/W/S share the row and city error position. F uses a newer registry row and a newly drawn error location. Section 6 shows additional identity fields.

2 Run and scoring specification

We specify the archived inputs, model settings, scoring rules and comparison units. The construction contract is in Construction and acceptance (Section 1).

2.1 Main input and output

The main direct-editing input contains 20 rows from different entities. Each original and its surrogates share 50 chunks per 1,000-row table. The zero-shot prompt asks for erroneous cells and correct values and asks the model to leave correct cells unchanged; replies are JSON. The aligned setting adds 20 labeled dirty/clean tuples and error positions. Whole-table methods additionally see repeated records of an entity.

2.2 Models, budgets and incomplete replies

Table S2: Archived model request and decoding settings.

Run family	Archived setting
DeepSeek-V4-Pro and Qwen3.8-Max	Thinking disabled; requested temperature 0
GLM-5.3	Reasoning effort low; thinking cannot be disabled; requested temperature 0
GPT-6.1-Sol	Reasoning effort low; client leaves sampling parameters at API defaults
Qwen2.5-7B/14B/32B-AWQ Instruct and Llama-3.1-8B-Instruct	Local vLLM and greedy decoding
Auxiliary OLMo-2-7B instruction/base variants	Local continuation probe; context limit 4,096 tokens

Replies are capped at 4,096 tokens, except 8,000 for GLM-5.3 and 1,500 for OLMo-2. The client retries replies marked incomplete at twice and four times the budget. Its heuristic checks the final delimiter, rather than an archived API termination reason. Eight of GLM-5.3's 500 Movies replies remain incomplete; all contain only `{"` and parse to no corrections. Their condition distribution and exclusion sensitivity are in Archived sensitivity checks (Section 7). Actual commercial flags are recorded in code/src/clients.py.

The scale statement records about 23,000 prompts: 9,600 zero-shot cleaning prompts, 600 aligned prompts, 3,216 conflict prompts and 9,600 continuation trials. The zero-shot count covers eight models and 24 tables, with 50 chunks per table. These are prompt counts, not a count of independent sampled entities.

2.3 Method information settings

Baran receives perfect detection where stated. Its aligned runs use the same chunks, labels and positions as DeepSeek-V4-Pro, averaged over three fixed seeds; whole-table Baran receives repeated records. Baran's Wikipedia-revision value models and Raha's knowledge-base detector are disabled. TabClean uses DeepSeek-V4-Pro, whole tables and 150 sampled development rows, which are excluded from scoring.

All twelve TabClean outputs on Hospital, Movies and Beers have the input shape and successful full-table execution logs. The development rows match `sample(n=150, random_state=42)` and are excluded from the reported scores. Section 6 reports repaired-cell counts and examples.

2.4 Cell-level scoring

Let E be the set of dirty/reference string differences and P the set of effective edits after no-ops are ignored. The set $R \subseteq P \cap E$ contains the cells set to the reference value. Repair precision is $p = |R|/|P|$, recall is $r = |R|/|E|$, and their harmonic mean is $F_1 = 2pr/(p+r)$. Detection replaces $|R|$ with $|P \cap E|$ in precision and recall.

The scorer sets precision to zero when $|P| = 0$ and recall to zero when $|E| = 0$. It sets $F_1 = 0$ when $p + r = 0$. The same zero-denominator conventions apply to detection. These rules also apply to bootstrap samples containing no errors or no proposed edits; such samples are retained, not dropped. The executable definition is code/src/probe_clean.py (`metrics`).

No-op and repaired-value comparisons ignore case and normalize whitespace, including spaces around commas. E uses original strings. Consequently, the five case-only title errors in each Movies condition belong to E but earn no repair credit under normalized comparison. This convention remains identical across conditions. Identity-column results and other-column results remain separate.

Repair intervals use 50,000 bootstrap reweightings. Hospital and Beers use independent mean-one, variance-one exponential weights for entities and replies; each row receives their product. Movies uses reply weights because each film appears in one reply. Counts are pooled with these weights before computing F1. O/W/S share corresponding entity and reply weights. F uses independent reply weights; on Hospital, the 33 institutions also present in O retain their shared entity weights without pairing different records. Section 7.9 gives the incidence structure and assumptions.

Two-sided tests compare the centered bootstrap deviation $|\Delta^* - \widehat{\Delta}|$ with $|\widehat{\Delta}|$, using $(1 + k)/(50{,}000 + 1)$ for k exceedances. Holm correction covers the 40 Hospital/Movies contrasts, with a separate eight-model family for Beers O–S. Significance means adjusted p below 0.05. Intervals use the 2.5th and 97.5th percentiles and are not multiplicity-adjusted. Repaired-cell counts and chunk wins/losses describe near-zero results. Hospital and Beers entities recur across chunks, so their wins/losses are not treated as independent sign trials. Movies sign tests use a separate eight-model Holm family. Seed 0 supplies the main tables; the three-seed minima and maxima describe construction variation rather than confidence intervals.

2.5 Row-continuation probe

The prompt supplies a header and ten consecutive rows and asks for the next row. It uses 100 trials per dataset for DeepSeek-V4-Pro and 50 for local models, clean or dirty context, with or without dataset names. The six datasets are Hospital, Movies, Beers, Rayyan, Flights and Tax; OLMo-2 omits Movies. The continuation probe covers DeepSeek, the four open-weight cleaners and the two OLMo-2 variants. Qwen3.8-Max, GLM-5.3 and GPT-6.1-Sol are outside this probe's archived coverage. Its absence of positive evidence concerns the tested models and settings.

Structural screen. A hard cell has a target value absent from the corresponding context column and missed by all four structural rules: column mode, most-frequent successor, approximate functional dependencies and previous-value ±1. For clean-context continuation the target is the clean next row; for dirty-context continuation it is the dirty next row. Strictly increasing unit-step counter columns, identified in the clean table, are omitted from cell scoring. This counter exclusion is separate from the ±1 rule applied to other numeric cells.

The rules use the complete source table, including records outside the ten-row prompt. Mode and successor queries subtract one vote for the queried target value; ties retain full-table first-occurrence order. For a dependency $d \to c$, the determinant has at most $N/2$ distinct values, and its modal mapping covers at least 99% of the N rows. Admission is determined on the full table. A successor query predicts d from its preceding value, then the admitted mapping predicts c; the target's mapping vote is subtracted when the predicted determinant equals its actual value. Dependency admission and tie order are not refitted after subtraction. These are offline structural exclusions, not an input-matched cleaner.

Parsing and leakage. Cell-accuracy rates condition on a parsed row. Of 9,600 stored trials, five calls failed, 511 other replies were unparseable, and 9,084 supplied a parsed row. These outcomes remain separate from cell denominators. A clean-value leakage opportunity has an erroneous dirty target whose clean reference is absent from the corresponding dirty-context column. The additional hard-leakage screen uses the full clean table and its clean predecessor. This oracle information may exclude values that the model cannot infer from its dirty prompt. Pooling across settings counts scored cell-trials, not necessarily distinct cells or independent observations.

Screened results. We apply the complete screen to the saved replies. Integer counts are stored in results/derived/uniform_continuation.json; code/src/uniform_continuation.py selects maxima and pools leakage from these counts. On Hospital, DeepSeek scores 221/468 (47%) on context-absent cells and 3/144 (2.1%) on hard

cells. Across tested settings, the largest hard-cell accuracy is 32/375 (0.085 after rounding). OLMo-2 base reaches a displayed maximum of 0.011. Hard clean-value leakage is 3/601 cell-trials for DeepSeek and 1/2,008 for the four open instruction cleaners.

Named and anonymous results are reported separately. Naming can change an individual rate by more than 0.02: Llama's Hospital clean rates are 3/72 and 1/72, a difference of 0.0278. Selecting erroneous target rows can also raise individual screened rates, as with Qwen-7B on Movies dirty (0.0070 to 0.0185). This latter setting changes the sampling pool rather than comparing identical target rows. The public-corpus infini-gram check finds Hospital fragments in a GitHub issue and tool documentation, and none of the other tables.

Row-held-out sensitivity. Removing the target row and both incident transitions, then refitting dependency admission and tie order, gives DeepSeek hard leakage of 3/607. Qwen-7B and Qwen-32B's error-row maxima become 0.028 rather than 0.029. The global maximum 32/375 and Hospital example 3/144 do not change. This separate calculation distinguishes full row-held-out fitting from the target-vote screen used for the reported results. Under the reported screen, the probe supplies no positive evidence attributable to benchmark-row memory beyond the excluded regularities; it does not estimate memory prevalence or rule out contamination.

2.6 Knowledge-conflict construction and matching

Targets have assigned attributes in both dirty and reference tables. Exact names, one-vowel-changed names and syllable controls share all other values, error positions and chunk rows. Hospital assignments are consistent across city/state/ZIP/county; Beers across city/state. Movies replaces old surrogate-director occurrences in Creator and Actors with the assigned director in all three versions, avoiding an old-director candidate in another column.

The fixed pool contains 57 films, 24 hospitals and 48 breweries that DeepSeek knows. Main chunks have three targets and 17 other-entity filler rows, with each target appearing once. A supplementary Hospital/Beers setting has three rows per entity, nine target rows and eleven fillers. Known subsets use model-specific factual queries: directors match a credited full name or a surname of at least four letters after accent/punctuation removal; locations require matching city and state code. Known subsets contain 52–57 films, 22–24 hospitals and 7–48 breweries.

The standard prompt asks for cleaning without changing correct cells. The table-evidence prompt adds: "Base every correction on evidence within this table only. Do not change a value just because it disagrees with what you know about real-world entities." Both prompts receive the same chunks and known subsets.

Director rewrites count outputs containing a full credited director name, including films with multiple directors. City rewrites require the normalized real city. The one-letter-name version instead measures restoration of the real name. Matching ignores case and normalizes whitespace; when a cell is listed repeatedly, its last value is scored. Directed matches and any edit to the watched cell are distinct indicators.

The rate first averages row indicators within an entity, then averages over its known entities. With s=1 it is matching films divided by known films; with s=3 it averages the three indicators per entity. Empty or unusable replies remain in the denominator. Controls are scored against the paired real fact with the same target subset. Intervals use 50,000 whole-reply resamples, pooling entity means and entity counts within each draw. Section 7.9 specifies the reply groups and fixed-context scope.

Executable provenance is in code/src/near_entities.py, the archived near-entity runs and the public core's anonymous movie indicators. The exact historical GPT screening dependencies and no-fallback behavior are specified in docs/gpt-screening.md and checked by tools/check_screening.py.

3 Rayyan retrieval and repair-credit criteria

The Rayyan audit compares saved PubMed fields with the benchmark's dirty and reference values.

3.1 Audit input and mechanical transformations

The audit uses 1,000 rows and eleven columns with 948 dirty/reference differences. All 722 date differences map `m/d/yy` to `d/int(yy)/mm`, with a two-digit input month in the final position: `1/1/13` to `1/13/01`, and `5/7/09` to `7/9/05`. Of 32 pagination differences, 31 turn ranges into month-name strings, such as `70-6` to `Jun-70`.

3.2 Title-based retrieval

Sampling draws up to 80 cells per error type with seed 0, including all 75 volume/issue, 32 pagination and twelve ISSN cells, for 279 sampled cells. Queries use the reference title, falling back to the dirty title when empty. The first eight ASCII-letter words longer than three characters are joined with AND in the title field; at least two words are required.

Among up to five candidates, the highest title similarity must reach 0.9. Similarity uses `SequenceMatcher` after NFC normalization, case folding, and replacing punctuation and underscores with spaces. There are 152 matched cells and 121 distinct PMIDs. These are automatically selected title matches, not manually certified whole-record matches.

3.3 Field checks and saved counts

Table S3: Rayyan field-check definitions, population/sample/match counts, and outcomes.

Check	Definition	Population / sampled / matched	Failed reference / passed dirty
Year	Compare the cell's extracted year with PubMed publication year; 00–26 become 2000–2026 and other two-digit years 1927–1999	722 / 80 / 44	43 / 42
Changed words	A nonempty set of reference-side changed words must all occur in the PubMed field's case-folded word set; accents remain	107 / 80 / 38	38 / unassessed
Volume/issue presence	Reference `-1` and dirty empty strings denote absence; compare field availability, not numerical equality	75 / 75 / 46	28 / 17
Pagination	Match the PubMed page string	32 / 32 / 18	18 / 17
ISSN	Accept print or electronic ISSN	12 / 12 / 6	6 / 0

The date field is `article_jcreated_at`; its exact dataset semantics are undocumented, so the test is publication-year consistency. Two reference volumes of `1` pass presence but disagree numerically with saved PubMed volumes `31` and `21` (PMIDs 26452069 and 25819808). Encoding was checked on the reference side only, hence the unassessed dirty marker.

For each of the five error-type strata h, let N_h be its population, m_h its title-matched sample and b_h the number of failed reference checks. We estimate the weighted failure rate as

$$\hat{\theta} = \sum_{h=1}^{5} \frac{N_h}{948} \frac{b_h}{m_h}.$$

The estimate is 95.2%, with interval 91.2%–97.8%. It summarizes failures under the audit input's 948-cell field composition; Table S3 gives the outcomes under each criterion. The 722-cell date stratum carries most weight. The 5,000 bootstrap draws resample saved binary outcomes within matched strata with seed 0 and fixed weights. The interval conditions on the selected matches and field criteria, without matching uncertainty or dependence between cells sharing a PMID.

Saved records include query titles, PMIDs, compared field values and outcomes, the publication year for dates, and a text prefix for encoding. Complete candidate responses, returned titles and similarity scores are not

retained. code/src/rayyan_audit.py specifies the query and checks; offline replay uses the saved matches and outcomes.

3.4 Repair credit and input versions

On the audit input, DeepSeek zero-shot output scores F1 0.002. At zero-based row 120, dirty ISSN `Mar-22` becomes output `0003-3022`, matching saved PMID 8916815, while the reference is `22-Mar`. This output gets no benchmark repair credit.

TabClean's distributed input has 960 differences. It removes commas, represents empty values as `empty`, and has twelve additional dirty-side error positions. Both inputs have 1,000 rows and eleven columns and share the date-conversion pattern. Scoring uses the actual TabClean pair, not the audit pair, and removes the 150 development rows sampled with `random_state=42`.

The remaining 850 rows have 808 errors. There are 498 effective edits and 461 credited repairs, so `F1=2*461/(498+808)=0.71` after rounding. Of the 461 credited repairs, 434 (94%) reproduce the reference date transformation.

The public entry is tools/check_rayyan.py; it binds audit and TabClean inputs, development indices and output to the recorded versions. See docs/rayyan-replay.md and Public verification (Section 5).

3.5 Other reference examples

Hospital assigns Helen Keller Memorial Hospital in Sheffield, Alabama to Jefferson County in 25 rows, although Sheffield is in Colbert County. Qwen3.8-Max, GLM-5.3 and GPT-6.1-Sol correct 10, 11 and 21 such cells, each scored as a false edit. Movies repeats the same writer in 68 reference cells, for example `Andrew Smith,Andrew Smith`.

4 Applying CoRe to another benchmark

Adapting CoRe starts with identifying what information the input provides for repairing each error.

4.1 Identity columns and recovery evidence

Identify columns whose values name or locate an entity, then profile their error kinds, strengths, positions and table-wide rules. Record the target convention separately from what the prompt actually states: a reference-side formatting convention is not necessarily an instruction given to the cleaner. For each error type, specify whether the input is a cell, a chunk, a chunk with labeled tuples and error positions, or the whole table.

R1 applies when a local conversion reaches the target under the stated convention. For R2, record candidates or regularities that narrow the choices, and whether they select a unique target. R3 records a required mapping or replacement value absent from the shown input. These descriptions need not be mutually exclusive: year candidates constrain the choice without deciding which source is preferred. A missing mapping can become available in another record or a labeled example. Classification follows the available evidence, not whether a particular model succeeds.

Table S4 covers the main error types. An ordinary name dictionary, an explicitly supplied candidate and the constructor's inventory are different sources of information. In particular, the syllable decoding analysis uses the constructor's inventory; the cleaner sees only the names in its chunk. We record observed input patterns separately from constraints used only by the decoding analysis.

Table S4: Recovery criteria for the principal error types. Candidate availability and exact target identification are recorded separately.

Error type	Evidence in the main input	Additional evidence or target convention
Hospital names and locations	R2 when another visible value or an observed word pattern constrains the typo; otherwise the exact replacement may remain R3	Same-entity records or labels may supply the value. Word knowledge is distinct from a candidate shown in the chunk.
Hospital numeric identity fields	Remaining digits constrain a corruption, but may not determine the missing digit	A visible matching value can help; random replacement numbers cannot be recovered from the real hospital's mapping.
Hospital syllable names	Names in the chunk expose a restricted pattern	Unique restoration under the constructor's inventory is an analysis-side R2 check, not an explicitly supplied cleaner resource.
Movies identifiers	A real slug can identify a familiar film; a fictitious slug does not determine a random reference ID (R3)	A mapping, retrieved value, labeled example or repeated record is needed for exact recovery.
Movies years	The range retains finite candidates (R2)	The middle rule is useful but not always correct; the target-source choice is not specified by the zero-shot prompt.
Movies titles and credits	Retained tokens, names in other columns or visible candidates can constrain edits (R2)	Omitted names and source-specific lists may still require absent values (R3).
Movies durations	An hour/minute string supports conversion, while an empty cell supplies no duration	Conversion is R1 only relative to a matching target convention; 50 conversions match the reference and 55 do not.
Movies ratings and genres	Dirty source values may constrain a replacement	Exact reference values can require a source mapping not present in the input (R3).
Movies numeric formatting	Separators and an added decimal suffix admit local conversion (R1)	The benchmark's target convention must be distinguished from an explicit prompt instruction.

Continued on next page

Table S4 (continued)

Error type	Evidence in the main input	Additional evidence or target convention
Beers city/state	The glued string retains both city and state (R1)	Splitting requires the table's column convention; the reference is not inferred from a known brewery.
Beers percentage suffix	The numeric value remains after removing the suffix (R1)	The target is the benchmark's fractional representation, not a newly inferred alcohol value.

4.2 Fresh records

Select a source of the same kind and map it to the benchmark schema and formatting. Document what has changed: records, entities, value distributions and error composition. Reapply error recipes to the new records and record cases that cannot transfer. Report paired and fresh-record comparisons separately, since fresh records may still describe familiar entities.

4.3 Two naming styles

Construct one-to-one English-word and syllable replacements. Retain generic terms and column conventions while avoiding real-value reuse, title-shape leakage and names that resemble the errors being tested. Keep non-identity values and error positions fixed in surrogates. Transfer the error mechanism rather than generating an unrelated corruption.

4.4 Acceptance and seeds

Encode structural, error-profile, naming/leakage and source-selection requirements as checks. Rebuild code-generated tables to test determinism and retain source snapshots for externally acquired fresh tables. Record exceptions explicitly. Use construction seeds to expose naming/error sensitivity, separating it from bootstrap variation over a fixed table. State which clean records are held fixed across seeds.

4.5 Fair comparisons and value preservation

Hold prompts, available labels, error positions and access to repeated records fixed within a comparison. Define cell-level scoring and the reference source. Add a separate continuation test with structural baselines when interpreting possible memorization. For conflicting facts, specify assigned values to preserve, pair name variants, identify each model's known subset, and measure directed factual rewrites separately from other edits.

4.6 Cost and scope

Most adaptation effort lies in error profiling and transferable recipes. Single-character typos require less source-specific handling than cross-source titles, credits, ratings and genres. A successful transfer needs a stated source convention and available recovery evidence, not just renamed entities. Matching and imputation are potential future settings, rather than tasks already validated by the present study.

Implementation details for the current study are in Construction (Section 1), Runs and scoring (Section 2) and Rayyan checks (Section 3).

5 Evidence and offline verification

The repository supports cell-level re-scoring where inputs and replies are available, and statistical replay from numerical witnesses where row-level materials are distributed separately. Table S5 lists the inputs, commands and outputs for each analysis.

5.1 Public offline entry

Use the dependencies in environment/requirements.txt. From the repository root, with a new output path:

```
python3 tools/core.py verify
python3 tools/core.py reproduce --output /new/public-results
python3 tools/check_rayyan.py
python3 tools/review_diagnostics.py
```

The scripts/verify.sh and scripts/reproduce.sh wrappers use the selected `PYTHON` interpreter. Offline replay requires no GPU, model key or network. It regenerates the condition contrasts, continuation summaries, conflict rates and Rayyan checks, including intervals and multiplicity corrections.

5.2 Manuscript to evidence map

All package paths below are relative to its extracted root. Generated paths are relative to the requested output directory.

Table S5: Evidence and executable replay routes by analysis.

Analysis and manuscript labels	Public evidence	Entry	Replayed products
Condition contrasts and continuation	Anonymous entity/reply count contributions in `data/witnesses/cluster_statistics.json`; archived counts and construction summaries; screened continuation counts	scripts/reproduce.sh or `tools/core.py reproduce`	Contrasts, cluster intervals, significance, seed/scale summaries, aligned results and figure CSVs. Continuation maxima and leakage use the screened integer counts.
Knowledge conflicts	Anonymous target indicators with reply membership, per-model known flags and directed/any-edit indicators	`tools/core.py verify` and public reproduce	Known-entity rates and reply-cluster intervals, `tables/near.tex`, `conflictcounts.tex` and the two conflict-figure CSVs; retains known denominators and control edits.
Reference quality and repair credit	Saved title matches and field checks, data/manifests/RAYYAN_REPLAY.json, `results/derived/rayyan_audit.json`, data/provenance/rayyan/, audit and actual TabClean inputs, development indices and archived output	tools/check_rayyan.py; public reproduce also invokes it	Weighted check-failure estimate/interval, `derived/rayyan_replay.json`, `tables/rayyan.tex`, Rayyan CSVs and the 808/498/461/434 scoring counts.

Movie witnesses contain numerical projections, not row-level movie values or replies. `CORE_WITNESSES.json` binds projections and `CORE_SOURCES.json` records historical local-validation dependencies. Rayyan replay uses saved matches and scalar checks, with saved encoding outcomes, rather than a fresh PubMed retrieval. Beers F provenance has the additional entry `python3 supplements/beers-f-input-provenance/verify.py`.

5.3 Figure and table generation

The aligned matrix gives all twelve method/dataset combinations, and the scale plot includes all eight cleaners. `tables/rayyan_checks.tex` combines sampling and title-match counts with the five matched-cell check fractions.

The first figure compares Movies identifier and year repairs using the counts in paper/latex/tables/moviescols.tex and the year CSV. Its generator is code/src/evidence_panels.py. The overview and Hospital construction figure

illustrate individual benchmark cells. Their generator is code/src/example_diagrams.py, called from code/src/figures.py, with PDF, PNG, SVG and source-cell CSV outputs. Section 6 extends the Hospital example.

Sources for the extended tables are `example.tex`, `checks.tex` and `aligned.tex`.

Table formatting is generated by code/src/paper_presentation.py. Figure 6 reads the 24-row aligned-results CSV through `figures.py`: DeepSeek's two series show 95% cluster intervals; Baran shows three-run means without intervals.

5.4 Authorized historical movie validation

Readers supplying the exact historical inputs, replies and scoring code can use:

```
python3 tools/core.py validate-private \
  --private-root /external/historical-root \
  --authorization-record /external/rights-record.json \
  --output /new/historical-validation-results
```

The authorization record contains `purpose`, `rights_basis`, a `scope` with `historical_inputs` and `historical_outputs`, and `redistribution: false`. The command checks 264 historical dependencies and 25 code files against the manifest before re-scoring saved direct-editing, aligned and conflict replies. The resulting witnesses must match the statistical core. Raw continuation scoring also requires the corresponding inputs. Historical re-scoring uses the archived records and replies, rather than new IMDb downloads or model calls.

5.5 Re-scoring, replay and missing records

Table S6 distinguishes checks on individual edits from calculations using saved counts and judgments.

Table S6: Evidence available for independent checking.

Level	Required materials	What can be checked
Cell-level re-scoring	Actual dirty/reference pair, saved reply, scoring code and development exclusions; exact historical Movies materials supplied separately	Reconstruct edits and compare each output with its reference; verify resulting counts and scores.
Statistical replay	Anonymous count contributions with entity/reply membership, entity rewrite indicators and matched-field outcomes	Recompute contrasts, cluster intervals and tests. Rayyan replay conditions on saved matches and checks rather than repeating retrieval.
Aggregate evidence	Reply-pattern summaries and token totals	Inspect reported aggregates; the omitted usage log and replies are required to reconstruct their individual entries.
Missing intermediate records	Twelve whole-table Baran runs lack correction lists; TabClean program bodies and complete PubMed candidate responses were not retained	Baran's count arithmetic, final TabClean outputs and selected PubMed checks remain available; the missing edits, programs and candidate-selection chain cannot be inspected from these records.

6 Extended displays and result details

This section gives the extended Hospital example, three-dataset condition overview, aligned/whole-table matrix, acceptance checks and Rayyan field-check display.

6.1 Three-dataset condition overview

The overview plot and plotted CSV give DeepSeek-V4-Pro's four-condition repair F1 on Hospital/Movies identity columns and all Beers columns. Whiskers give 95% cluster intervals. O/W/S are paired; F uses fresh real records. Beers F is descriptive and has no interval.

Main-paper Figure 1 reports repaired/error counts for Movies identifiers and years (Table 10). Figure 4 shows the year-choice distribution; the middle-year rule matches 167/190 references and was not supplied to the zero-shot cleaner.

6.2 Hospital row across four conditions

Table S7 shows Hospital identity fields across conditions, with corrupted cities in bold. O/W/S share a row and error position. F draws a newer registry row with updated quality measures rather than a row-paired replacement of O.

Table S7: Hospital identity fields across the four conditions; bold cities are corrupted.

Version	Name	Address	City	County	Phone
O	callahan eye foundation hospital	1720 university blvd	**birminghxm**	jefferson	2053258100
F	southeast health medical center	1108 ross clark circle	**dxthan**	houston	3347938701
W	timber hollow regional hospital	5749 glen road	**xill pine**	sunset prairie	6175712390
S	vasrhokel regional hospital	8928 zarquen street	**hepmxrmxr**	lotorcal	6061657772

Main-paper Figure 3 focuses on hospital names and the clean-to-dirty city transfer, with F separated from paired O/W/S. Its reference/dirty city pairs are `birmingham`/`birminghxm`, `mill pine`/`xill pine`, `hepmormor`/`hepmxrmxr`, and `dothan`/`dxthan`, respectively. Hospital F contains 34 hospitals, 33 shared with O, with updated quality measures and newly drawn error locations.

6.3 Acceptance checks

The 506 checks cover 21 tables.

Table S8: Construction acceptance checks by category.

Category	Checks	What is checked
Structure	67	Shape and columns; error cells where the original has them; identical non-identity cells; different seeds with shared fresh clean tables.
Error kinds and counts	137	Per-column kinds and counts as in the original; formatting rules; errors only in rows that carry them.
Error strength	15	Characters replaced per error cell within the tolerance of the original.
Naming and leakage	263	One fictitious value per real value and back; no real value reused; no word of a real title; title shapes no closer to real ones than chance; names that look like names.
Fresh entities	6	No benchmark entity where new entities are required; locations checked against sources.
Reproducibility	18	Every table rebuilt byte for byte under two hash seeds.

The checks follow each family's fresh-source design: Hospital F uses newer rows largely about the same institutions. The tables pass 505/506 checks. Hospital W at seed 0 has a maximum error-strength difference of 10.5 percentage points, exceeding the eight-point tolerance by 2.5 points. Seeds 1 and 2 pass, with all error positions

unchanged. Construction and acceptance (Section 1) gives the thresholds; Sensitivity checks (Section 7) gives the passing-seed contrasts.

6.4 Aligned and whole-table repair

Identity-column repair F1 is shown below in full and in main-paper Table 13. Aligned methods receive the same chunks, 20 labeled tuples and error positions. Baran is the mean of three runs; whole-table Baran uses perfect detection. TabClean uses DeepSeek-V4-Pro with development rows excluded.

Table S9: Identity-column repair F1 under aligned and whole-table information.

Dataset	Method	O	F	W	S
Hospital	DeepSeek, aligned	0.97	0.98	0.76	0.75
Hospital	Baran, aligned	0.29	0.32	0.27	0.28
Hospital	Baran, whole table	0.91	0.93	0.93	0.93
Hospital	TabClean	1.00	1.00	0.96	0.99
Movies	DeepSeek, aligned	0.64	0.67	0.09	0.07
Movies	Baran, aligned	0.00	0.00	0.00	0.00
Movies	Baran, whole table	0.27	0.26	0.26	0.25
Movies	TabClean	0.00	0.00	0.03	0.00
Beers	DeepSeek, aligned	1.00	1.00	1.00	1.00
Beers	Baran, aligned	0.19	0.77	0.20	0.22
Beers	Baran, whole table	0.94	1.00	0.94	0.96
Beers	TabClean	1.00	1.00	1.00	1.00

Main-paper Figure 6 displays Hospital/Movies zero-shot and aligned results from all 24 rows of the plotted CSV. DeepSeek's zero-shot and aligned series show 95% cluster intervals. Baran's aligned series shows the mean of three runs, without error bars. Table 13 includes Beers and the whole-table settings. All three datasets use identity-column scoring here. The plotting implementation is code/src/figures.py (`fig3`).

Whole-table Baran and TabClean combine high scores with little change under renaming on Hospital and Beers. Movies has a different profile: whole-table Baran remains at 0.25–0.27, while TabClean is at 0.00–0.03. The latter's small variation accompanies almost no identity repairs, rather than successful recovery across names. Aligned Baran also remains at zero on Movies.

6.5 Rayyan field-check display

Main-paper Table 17 reports population, sample and title-match counts with reference-failure and dirty-pass fractions. The field-check plot shows these fractions among matched cells. The count table, count CSV and display CSV provide their numerical values. The changed-words dirty check is unassessed, not zero.

The field definitions and conditional weighted estimate are specified in Rayyan retrieval and scoring details (Section 3). The plot and table use the same five strata and matched cells.

6.6 Full model comparison

The eight-model Hospital plot and plotted CSV compare the four conditions with 95% cluster intervals. Across the three Qwen2.5 sizes, the O–S repair-F1 difference grows while repair on S remains low.

6.7 TabClean: archived output evidence

The twelve TabClean condition outputs have successful full-table execution logs in artifacts/raw_outputs/e4/. The 150 development rows match `sample(n=150, random_state=42)` and are excluded from the scores. On Movies, the original, fresh, word and syllable conditions repair 0, 0, 5 and 0 identity cells, respectively, out of 319, 313, 319 and 319 errors after exclusion. These counts yield the near-zero identity-column F1 values.

For a Beers example outside the development rows, `San Francisco CA` in City and an empty State become `San Francisco` and `CA`, matching the reference. Both pieces of information are present in the dirty input. All 104 non-development city-error rows in Beers O have the correct city and state in the final output. code/src/e4.py scores these outputs after development-row exclusion.

6.8 Extended figures

Figures S3–S7 show the condition overview, year candidates, model comparison, aligned repair and Rayyan field checks.

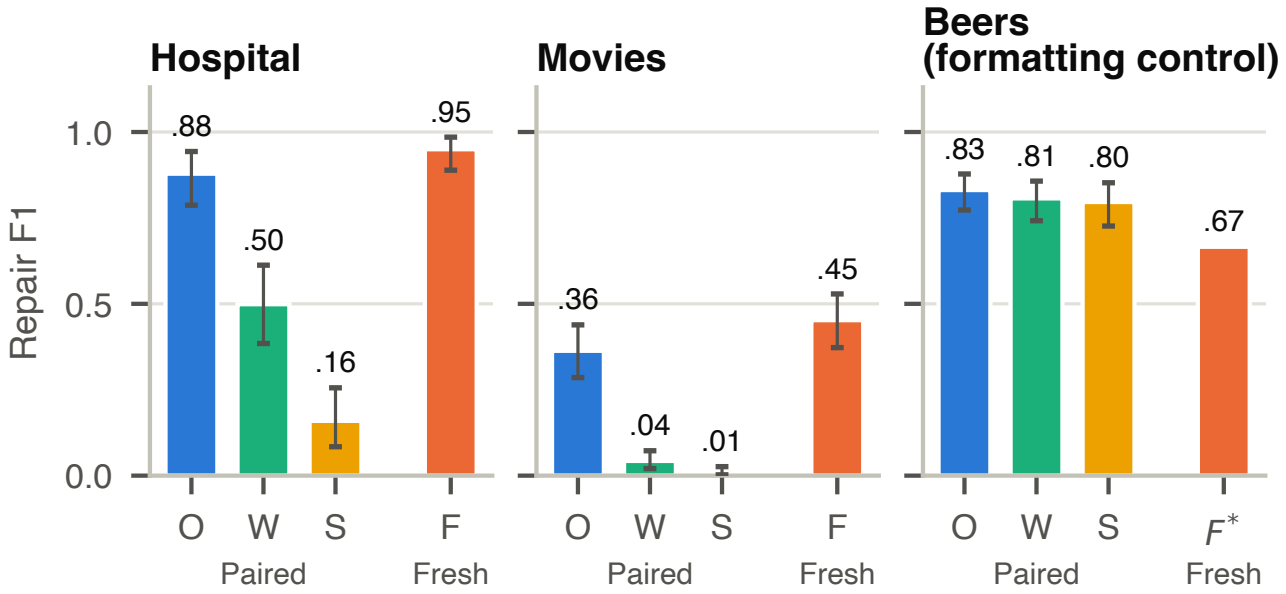


Figure S3: Four-condition DeepSeek-V4-Pro repair F1 on Hospital/Movies identity columns and all Beers columns. Whiskers give 95% cluster intervals. F is unpaired; Beers F is descriptive, without an interval.

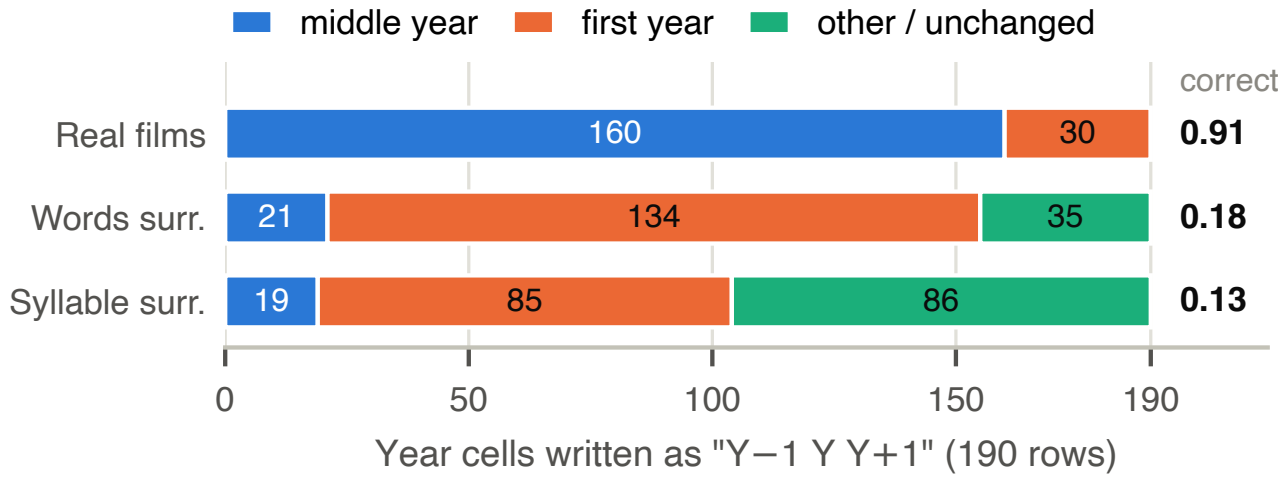


Figure S4: Movies year candidates and repairs. Taking the middle candidate matches 167/190 references; the cleaner was not supplied this rule.

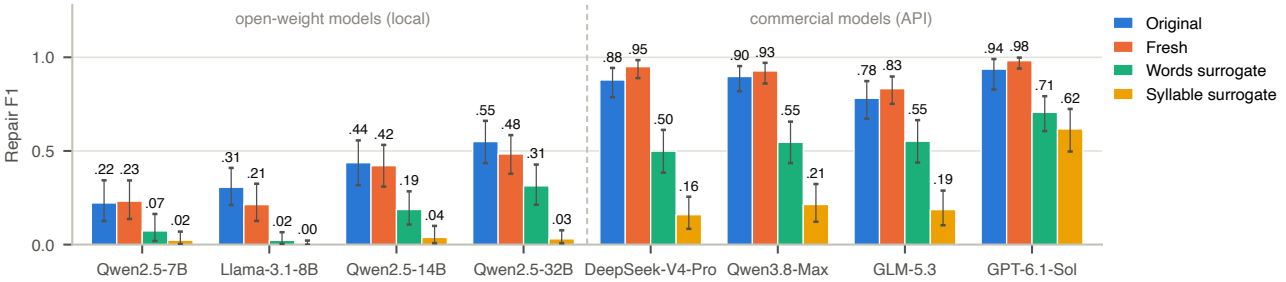


Figure S5: Full eight-model Hospital identity-column comparison with 95% cluster intervals. The scale trend concerns the tested Qwen2.5 family on Hospital.

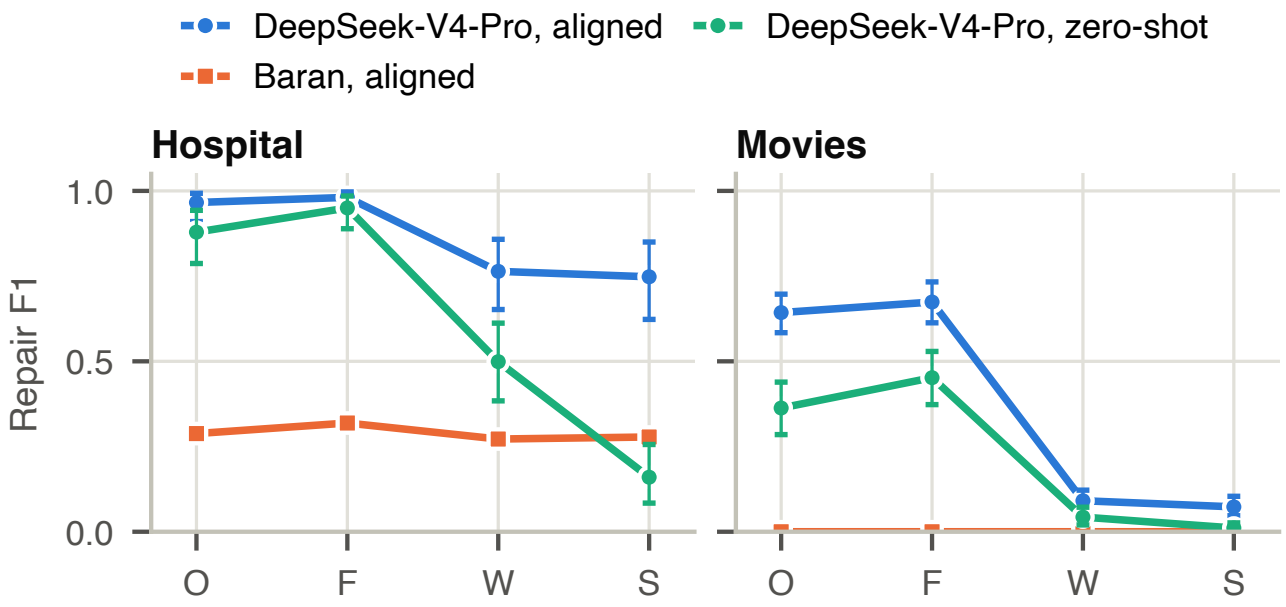


Figure S6: Zero-shot and aligned identity-column repair F1. DeepSeek's two series show 95% cluster intervals; aligned Baran shows three-run means without intervals.

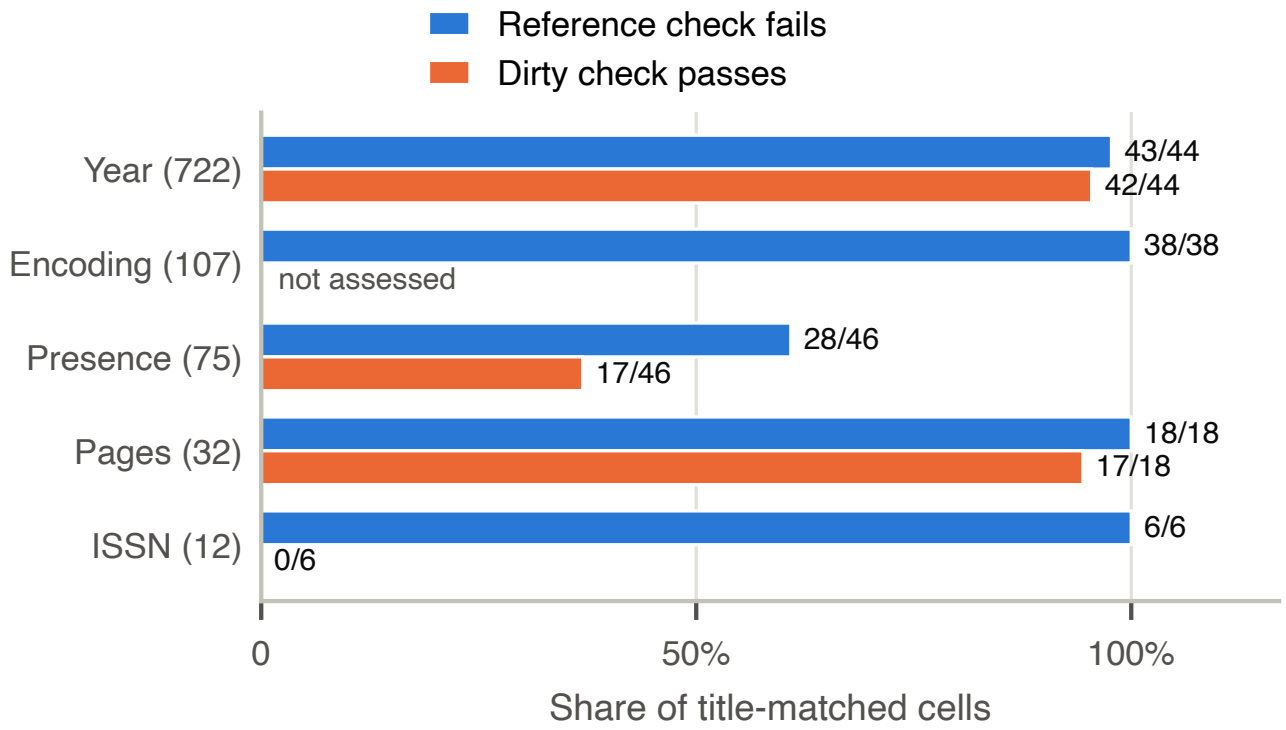


Figure S7: Rayyan stratified field checks using the saved matches and criteria in Section 3. The dirty changed-word check is unassessed.

7 Archived sensitivity checks and output patterns

We examine construction strength, incomplete replies, formatting repairs and title-match coverage using the saved tables, replies and retrieval records.

7.1 Hospital construction strength

Hospital's identity columns contain 219 erroneous cells. The original and all three word-based surrogate tables have identical error positions. The acceptance check groups equal-length corruptions by the number of replaced characters and permits an eight-percentage-point difference in each bin.

Table S10: Hospital character-replacement strength distributions and acceptance outcomes.

Table	One character	Two characters	Three or more	Maximum difference from O	Strength check
O	118 (53.9%)	64 (29.2%)	37 (16.9%)	—	Reference
W, seed 0	136 (62.1%)	69 (31.5%)	14 (6.4%)	10.5023 points	Outside tolerance
W, seed 1	123 (56.2%)	66 (30.1%)	30 (13.7%)	3.1963 points	Pass
W, seed 2	118 (53.9%)	74 (33.8%)	27 (12.3%)	4.5662 points	Pass

Seed 0's largest difference is 10.5023 points, or 2.5023 points beyond the tolerance. Character counts describe the corruption distribution; they do not measure a cleaner's difficulty. The following comparison uses the two passing word-based surrogate constructions, alongside the corresponding fresh and syllable constructions. Ranges span seeds 1 and 2, rather than bootstrap samples. Differences use the archived scorer's four-decimal condition scores, before rounding for display in the main tables.

Table S11: Hospital condition-contrast ranges over passing word-based surrogate seeds 1 and 2.

Cleaner	O-W	F-W	W-S	O-S
DeepSeek-V4-Pro	.3166–.3303	.3395–.3690	.3331–.3604	.6634–.6770
Qwen3.8-Max	.3102–.3192	.3372–.3444	.3628–.3814	.6820–.6916
GLM-5.3	.2153–.2623	.2764–.2865	.3436–.3555	.5708–.6059
GPT-6.1-Sol	.2333–.2406	.2764–.2919	.0993–.1048	.3381–.3399
Qwen2.5-7B	.1335–.1712	.1876–.2197	.0222–.0664	.1934–.1999
Llama-3.1-8B	.2511–.2921	.1364–.2360	.0098–.0440	.2951–.3019
Qwen2.5-14B	.2517–.2576	.3057–.3141	.1169–.1669	.3745–.4186
Qwen2.5-32B	.2115–.2466	.2366–.2996	.2698–.2804	.4919–.5164

All eight cleaners retain positive F-W and W-S contrasts in the two passing constructions. O-S does not involve W.

7.2 Incomplete Movies replies

GLM-5.3 has eight incomplete replies among 500 chunks over ten Movies tables: O and three seeds each of F, W and S. Six occur in the four seed-0 tables. The table includes zero counts to make that scope explicit; chunk numbers are one-based within a table.

Table S12: GLM-5.3 incomplete Movies replies across the ten archived tables.

Condition	Incomplete / 50	Chunk numbers
O	1	24
F, seed 0	0	—
F, seed 1	0	—
F, seed 2	0	—
W, seed 0	1	47
W, seed 1	0	—
W, seed 2	0	—
S, seed 0	4	14, 16, 26, 47
S, seed 1	0	—
S, seed 2	2	14, 47

Every incomplete reply contains only `{"`, yielding no parsed corrections. Incompleteness is identified by the archived client's delimiter heuristic, not an archived provider termination reason. The main scores count these replies as no corrections.

For the exclusion check, O-W removes a paired chunk if either reply is incomplete, retaining 48 common pairs. F-W removes incomplete replies independently, retaining 50 fresh chunks and 49 word-based surrogate chunks. Both analyses use 50,000 reply reweightings with the same pairing rules and the full 40-comparison Holm family (eight cleaners × five Hospital/Movies contrasts).

Table S13: GLM-5.3 Movies contrasts using all replies or complete replies.

GLM Movies comparison	Replies included	Chunks, left/right	Difference	95% percentile interval
F-W	All	50 / 50	.1660	[.1056, .2278]
F-W	Complete only	50 / 49	.1654	[.1040, .2282]
O-W	All	50 / 50	.0698	[.0320, .1111]
O-W	Complete pairs	48 / 48	.0680	[.0292, .1096]

Both contrasts are significant under the 40-comparison Holm procedure. F-W has adjusted p below 0.001 with either selection; O-W has p=0.022 using all replies and p=0.029 using complete pairs. The intervals are unadjusted percentile intervals.

7.3 Rating-count repairs

DeepSeek-V4-Pro's Movies replies show a different pattern on rating counts than on years. The dirty and reference rating-count columns are identical across O/W/S. Repairs of these counts are all exact comma deletions, but the model applies the transformation unevenly.

Table S14: DeepSeek-V4-Pro Movies rating-count proposals, effective edits, and repairs.

Condition	Error cells	Raw proposals	No-op proposals	Effective edits	Repairs	Wrong edits on error cells	Correct cells modified	Error cells without effective edits
O	768	80	73	7	2	1	4	765
F	1,000	69	43	26	26	0	0	974
W	768	170	117	53	24	8	21	736
S	768	313	190	123	90	16	17	662

All listed proposals have valid row and column addresses. No-op proposals equal the dirty value under the scorer's case/whitespace normalization and do not count as edits. Repeated proposals retain the last effective edit; a later no-op does not erase it.

The 90 S repairs occur in seven chunks: 2 (18 repairs), 15 (18), 21 (8), 31 (15), 41 (15), 48 (1), and 50 (15). O repairs occur in two chunks, W repairs in five and F repairs in three.

7.4 Rayyan title-matching coverage

The 279 sampled cells yield 152 title-matched cells (152/279 = 54.5%), associated with 121 distinct PMIDs. Field-specific match rates and checks are as follows. A dash means that the dirty field was not assessed, rather than a failed check.

Table S15: Rayyan cell-level title-matching coverage and matched field outcomes.

Field type	Sampled	Matched	Match rate	Matched dirty fields passing	Matched reference fields passing
Date	80	44	55.0%	42 / 44	1 / 44
Encoding	80	38	47.5%	—	0 / 38
Missing volume/issue	75	46	61.3%	17 / 46	18 / 46
Pagination	32	18	56.3%	17 / 18	0 / 18
ISSN	12	6	50.0%	0 / 6	0 / 6

Dates are checked against publication year, volume/issue against presence, pagination against the page string, ISSN against print or electronic identifiers, and encoding against the changed-word criterion. The weighted estimate uses matched cells within each type and the 948-cell type composition for weights. Unmatched cells are unassessed. Rayyan audit criteria (Section 3) specifies the field semantics and conditional interval.

7.5 Exact direct-editing prompt

The following literals come from the archived cleaning source. They specify the zero-shot direct-editing task, not the continuation or conflict prompts. Python's doubled braces escape literal JSON braces; `{block}` is replaced by the input CSV, including its `_row` column.

System message:

```
You are a meticulous data cleaning assistant. Reply with JSON only.
```

User-message template:

```
The CSV table below contains some erroneous cells (for example typos or corrupted values). The first
column `_row` numbers the rows.
Find the erroneous cells and give their correct values. Only list cells you believe are wrong; do
not rewrite correct cells.
Reply with JSON: {{"corrections": [{{"row": <_row>, "column": "<column name>", "value": "<corrected
value>"}}]}}

{block}
```

7.6 Recorded token usage

The historical usage log contains 11,413 entries, including entries not assigned to the paper's final workloads. Its input and output totals are reported separately below. Reasoning tokens are already included in output tokens and must not be added again.

Table S16: Historical usage-log totals; reasoning tokens are included in output tokens.

Logged model	Entries	Input tokens	Output tokens	Known reasoning tokens within output	Entries with unknown reasoning count
DeepSeek-V4-Pro	3,169	7,512,112	2,615,556	0	0
Qwen3.8-Max	2,953	4,931,264	654,694	0	0
GLM-5.3	2,806	3,971,489	2,984,507	2,565,488	0
Gemini-3.5-Flash	12	23,830	3,240	Unknown	12
GPT-6.1-Sol	2,473	3,320,127	588,719	295,011	0
Total	11,413	19,758,822	6,846,716	2,860,499 known	12

The log includes retries, pilots and Gemini usage outside the eight-cleaner evaluation. Its attribution to final experiments and coverage of local or cached runs are incomplete, so these totals cannot establish the study's total tokens or monetary cost. Run and scoring specification (Section 2) gives the request settings and approximately 23,000-prompt workload.

7.7 Diagnostic calculations

The repository's tools/review_diagnostics.py recomputes construction, block-count and saved-match checks. Its output identifies aggregate-only reply-pattern and usage evidence separately. docs/reproduction.md lists the required inputs.

From the repository root:

```
python3 tools/review_diagnostics.py
```

The calculations use saved observations and do not invoke model generation. Recomputing reply-level diagnostics requires the corresponding archived inputs and replies; usage totals require the historical usage log.

7.8 Hospital relationship and edit audit

The Hospital audit reads the ten actual dirty/reference pairs and the eight cleaners' saved 20-row spread-chunk replies. It independently parses edits, removes normalized no-ops and checks per-chunk error, prediction, detection and repair counts. Repeated proposals retain the last effective edit, as in the reported scoring. The entry is `supplements/hospital-semantics/audit.py`:

```
python3 supplements/hospital-semantics/audit.py \
  --repo-root . --output /new/hospital-audit
```

The outputs give field mappings, effective edits with reply/chunk/row addresses, uncredited geographic edits, per-column counts and source hashes. They permit inspection of the edits summarized in Table S1; they do not estimate a causal effect of cross-field geography or replace the original scores. No model or retrieval request is made.

7.9 Shared entities and reply contexts

Hospital's 50 chunks each contain 20 different hospitals. Across chunks, an original hospital appears in 1–28 replies; the 34 fresh hospitals each appear in 11–30. The original and surrogate inputs have 45 hospitals and identical chunk assignments. Their entity–reply incidence graphs are connected, so connected components do not provide a collection of independent groups for resampling. This overlap is distinct from the pairing used to compare O/W/S and from Holm's correction over comparisons.

We check concentration in the fixed Hospital outputs by omitting one hospital's rows and recomputing repair F1 from the remaining error, edit and repair counts. Paired contrasts omit the same original hospital from both

conditions. Fresh comparisons omit one hospital from one side at a time, leaving the other side fixed. All O–W, W–S, O–S and F–W differences remain positive for every tested omission in each of the three constructions. Table S17 gives the ranges for the main reported contrasts. The small auxiliary O–F difference for Qwen2.5-14B at seed 0 crosses zero, from a full-output difference of 0.0160 to a deletion range of −0.0008–0.0355. This check measures the composition of saved outputs; it neither regenerates replies with a hospital removed nor estimates a confidence interval.

Table S17: Hospital repair-F1 differences after omitting one hospital. Entries are minima and maxima across the three seed-specific deletion checks, not confidence intervals.

Model	W–S	O–S	F–W
DeepSeek-V4-Pro	.3215–.3721	.6506–.7374	.3260–.4634
Qwen3.8-Max	.3177–.3958	.6706–.7171	.3260–.3917
GLM-5.3	.3279–.3812	.5579–.6264	.2598–.3020
GPT-6.1-Sol	.0736–.1149	.3037–.3882	.2614–.3006
Qwen2.5-7B	.0097–.0738	.1805–.2129	.1421–.2331
Llama-3.1-8B	.0052–.0490	.2829–.3117	.1217–.2451
Qwen2.5-14B	.1054–.1768	.3560–.4376	.2182–.3291
Qwen2.5-32B	.2535–.2962	.4819–.5324	.1590–.3188

In the conflict tests, every known target appears in one reply per setting, with one or three rows in that reply. Known targets occupy 8 replies for Hospital, 6–16 for Beers and 19 for Movies, with 1–3 known entities per reply. Conflict intervals sample complete replies, carrying all selected entities and their within-entity row means. Each draw divides the sum of these means by the number of sampled entities, rather than averaging reply-level rates. This retains equal entity weights when the known subset gives replies different numbers of targets. All-or-none outcomes are reported as counts and rates, without a confidence interval inferred from a constant empirical distribution.

The cluster model conditions on the constructed tables, selected known entities and background values. It allows dependence within an entity or reply and treats units sharing neither cluster as independent. Intervals therefore concern these fixed constructions, rather than new backgrounds, regrouped prompts or repeated API calls. The multiplicative weights follow Owen and Eckles's crossed-factor bootstrap. The Hospital omission ranges in Table S17 describe a separate composition check.

The offline command `python` tools/replay_cluster_statistics.py reads data/witnesses/cluster_statistics.json, containing anonymous entity/reply indices, integer count contributions and target indicators. It calculates the 40-comparison and Beers families, conflict intervals and aligned comparisons. Beers F is descriptive because its entity composition and error distribution differ from O/W/S. The three Beers F runs identified in the provenance supplement are scored against their hash-bound historical inputs; every archived chunk count is reproduced.

7.10 Year choices within reference groups

Of Movies' 190 year ranges, 167 have the middle candidate as their reference, 22 have the first and one the last. Table S18 separates the middle-reference cases from the other 23. DeepSeek-V4-Pro repairs 155/167 middle-reference cases on O, compared with 19/167 on W and 17/167 on S. On the other cases it repairs 18/23, 16/23 and 8/23, respectively. In the paired O–W replies, 108 of the 160 middle-year choices on O become first-year choices on W. The decline thus includes changes within the same candidate sets, rather than only the 23 cases in which the middle rule disagrees with the reference.

Table S18: DeepSeek-V4-Pro's Movies year choices by reference group. Other is an output outside the three candidates; unchanged retains the dirty range. No output selects the last candidate.

Reference group	Condition	Correct / cases	First	Middle	Other	Unchanged
Middle	O	155/167	12	155	0	0
Middle	W	19/167	118	19	30	0
Middle	S	17/167	77	17	71	2
First or last	O	18/23	18	5	0	0
First or last	W	16/23	16	2	5	0
First or last	S	8/23	8	2	13	0

These counts sum to the 173, 35 and 25 repairs and the choice totals in main-paper Figure 4. The prompt does not specify a candidate-selection rule; the grouping describes agreement with the benchmark's reference rather than execution of an instructed middle-year rule. The diagnostic entry is `supplements/dependence-and-year/diagnostics.py`; it reads the archived inputs and replies, with historical movie members checked against `CORE_SOURCES.json`. It exports the incidence, omission and paired-year tables without model calls or new interval estimates.